

Undergraduate Medicine

Endocrinology Study Pack

CharaNas Medicine Bot — Specialty Notes

Expanded notes with original schematic figures for revision. These are educational drawings inspired by standard undergraduate teaching themes, not copied textbook plates. Always follow your faculty curriculum and latest guidelines.

Section	Focus
1	Hormonal axes overview
2	Diabetes emergencies
3	Thyroid disease
4	Adrenal, calcium, pituitary
5	Checklist & books
6	Quick pearls from the bot notes
7	Recommended book sources

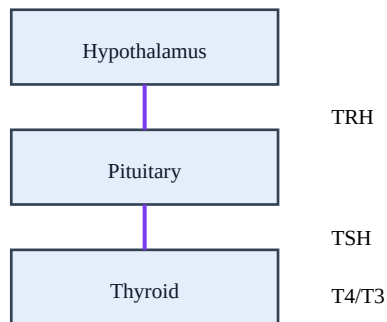
1. Hormonal axes overview

Endocrine diagnosis uses clinical features plus paired hormone levels (e.g., TSH with free T4). Feedback loops explain many patterns.

Diabetes mellitus: type 1 (insulin deficiency, DKA risk) vs type 2 (insulin resistance $\pm$ deficiency). Diagnose with fasting glucose, HbA1c, or OGTT per criteria.

Always interpret labs with drugs, illness, and assay caveats in mind.

Figure: Hypothalamic–pituitary–thyroid axis



Feedback loops regulate TSH; interpret TFT patterns clinically

Schematic figure for learning — verify details in your recommended textbooks.

Revision prompts

- Explain this section aloud in 60 seconds as if teaching a junior student about endocrinology.
- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

2. Diabetes emergencies

DKA: hyperglycemia + ketosis + metabolic acidosis. Priorities: IV fluids, insulin, potassium monitoring/replacement. Do not start insulin if severe hypokalemia is uncorrected.

HHS: marked hyperglycemia/hyperosmolarity, usually older T2DM, milder ketosis — careful fluid/electrolyte correction.

Hypoglycemia: Whipple triad idea — low glucose, symptoms, resolution with glucose. Give fast carbohydrate or IV/IM glucagon pathways as needed.

Revision prompts

- Explain this section aloud in 60 seconds as if teaching a junior student about endocrinology.
- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

3. Thyroid disease

Hypothyroidism (often Hashimoto in iodine-sufficient areas): fatigue, weight gain, cold intolerance, bradycardia — high TSH, low free T4 in primary disease.

Hyperthyroidism/Graves: heat intolerance, weight loss, tremor, tachycardia, goiter ± orbitopathy — low TSH, high free T4/T3.

Thyroid storm and myxedema coma are ICU-level emergencies.

Revision prompts

- Explain this section aloud in 60 seconds as if teaching a junior student about endocrinology.
- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

4. Adrenal, calcium, pituitary

Adrenal insufficiency crisis: shock, hyponatremia, hyperkalemia — give stress-dose steroids and fluids urgently while investigating.

Hypercalcemia: 'stones, bones, groans, psychiatric overtones'; ECG QT changes; treat severe cases urgently. Hypocalcemia: tetany, Chvostek/Trousseau.

Pituitary masses may present with headache/visual field defects (bitemporal hemianopia) and hormone excess/deficiency.

Revision prompts

- Explain this section aloud in 60 seconds as if teaching a junior student about endocrinology.
- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

5. Checklist & books

Checklist: glucose/ketones, TFTs, electrolytes/calcium, medication list (steroids, amiodarone, lithium), sick-day rules teaching for adrenal/diabetes patients.

Books: Williams Textbook of Endocrinology; Greenspan; Harrison's endocrine chapters.

Revision prompts

- Explain this section aloud in 60 seconds as if teaching a junior student about endocrinology.
- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

6. Quick pearls

High-yield reminders packaged with this specialty in the CharaNas bot:

1. DKA: hyperglycemia + ketones + acidosis; fluids, insulin, K+.
2. Hashimoto → hypothyroidism; Graves → hyperthyroidism.
3. Metformin first-line for many T2DM patients if eGFR allows.
4. Hypocalcemia: Chvostek/Trousseau; check Mg and PTH.
5. Adrenal crisis: shock + hyponatremia/hyperkalemia — give steroids/fluids urgently.
 - Link symptoms → anatomy/physiology → investigation → first management step.
 - Write one OSCE-style explanation for a common presentation in this specialty.
 - After reading, do the bot Short MCQ and Case-based Question for active recall.

7. Recommended book sources

Standard undergraduate references commonly used internationally. Use the edition your college recommends. Figures in commercial textbooks are copyrighted; use library/licensed access for full plates and photographs.

#	Reference
1	Williams Textbook of Endocrinology
2	Harrison's — Endocrinology
3	Greenspan's Basic & Clinical Endocrinology

How to study with this PDF: read one section → sketch the figure from memory → answer bot MCQs/cases → review mistakes → revisit the matching section.

Disclaimer: educational aid only; not a substitute for clinical guidelines, faculty teaching, or licensed textbooks.